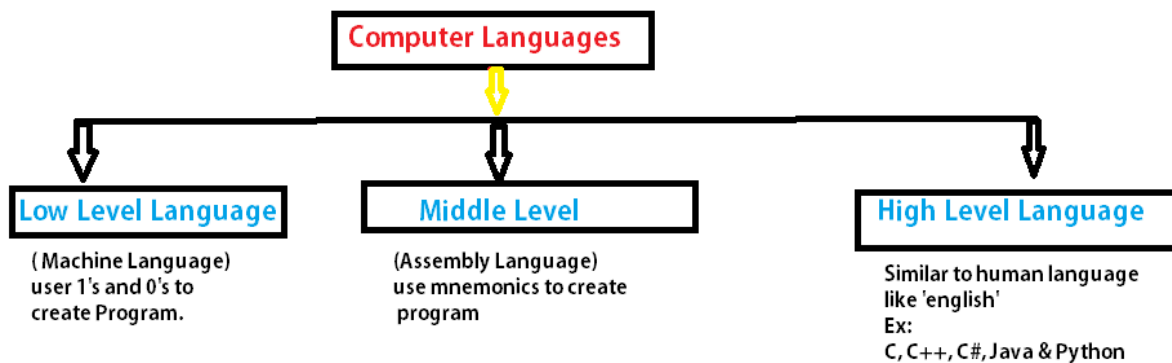


Java

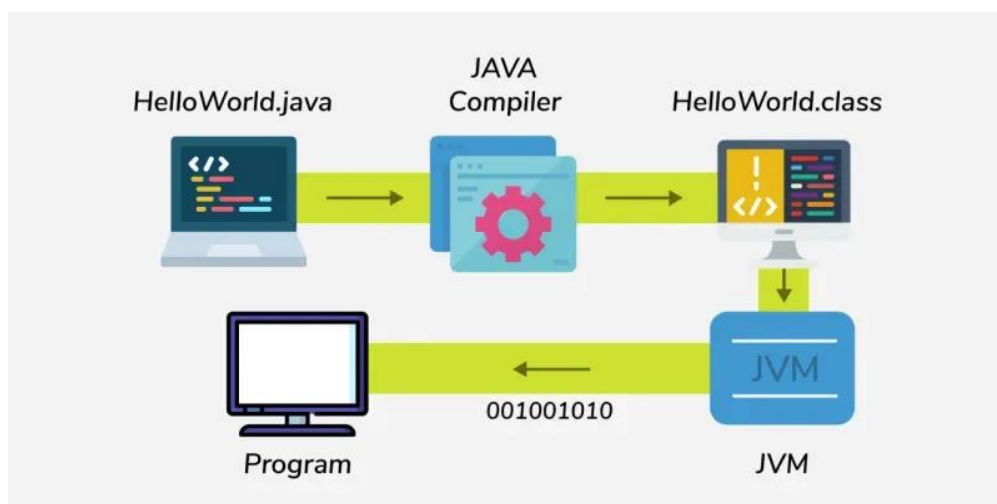
What is Programming Language ?

- A programming language is a way to communicate with computers.
- It consists of a set of rules, syntax, and instructions.
- Programmers use it to write programs (software, apps, systems).
- It converts human logic into a form that the computer can understand.
- Programming languages are mainly divided into low-level and high-level.



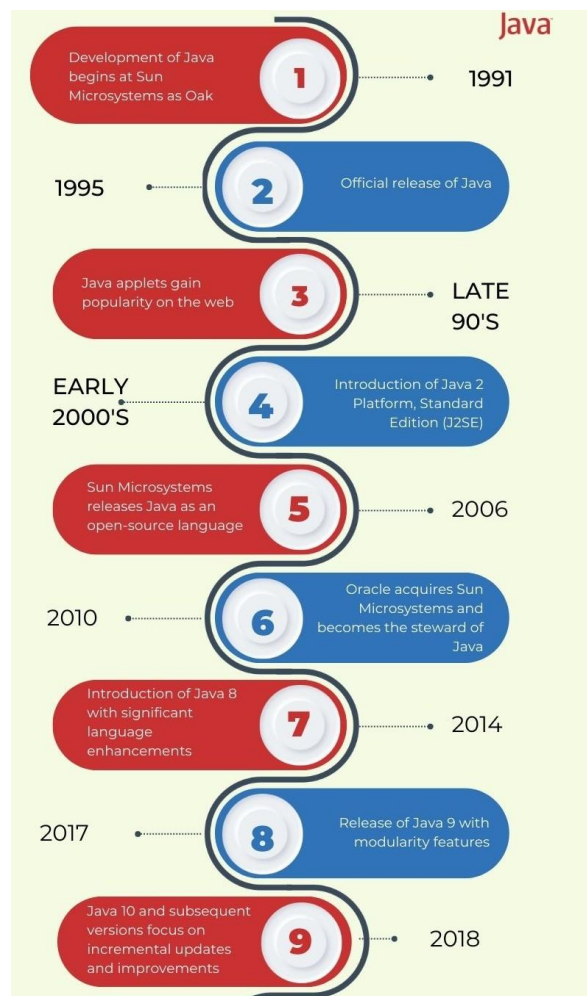
Introduction of Java :-

- Java is a high-level, object-oriented programming language.
- It was developed by James Gosling at Sun Microsystems in 1995.
- Java is designed with the principle of “Write Once, Run Anywhere” (WORA).



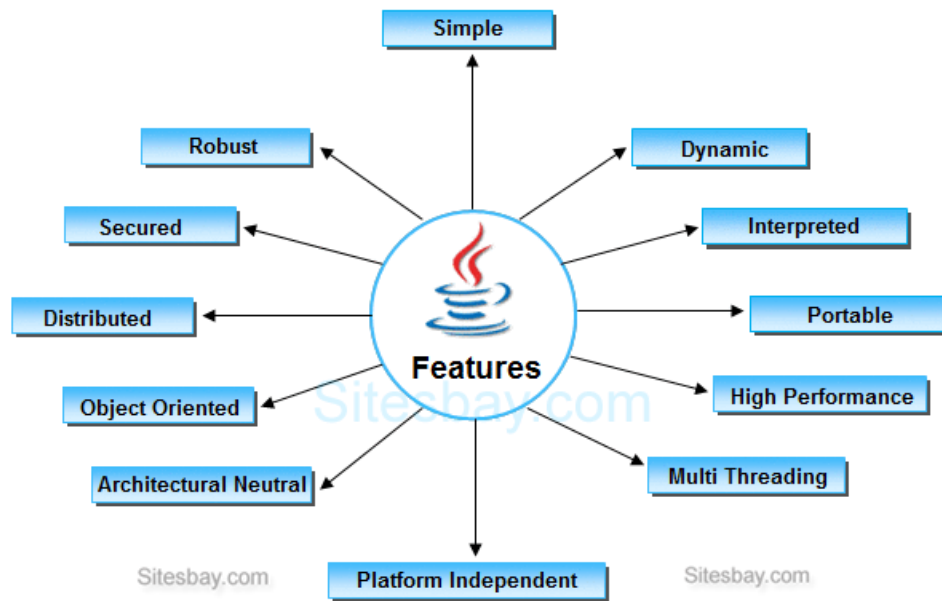
History of Java :-

- Java was developed in 1991 by James Gosling and his team at Sun Microsystems.
- It was first named Oak, later renamed to Java in 1995.
- Java was officially released as Java 1.0 in 1995.
- Its slogan was “Write Once, Run Anywhere” (WORA) using the JVM.
- In the late 1990s, Java became popular for web applications and applets.
- In 2010, Oracle Corporation acquired Java from Sun Microsystems.



Features of Java :-

Java is a high-level, object-oriented, platform-independent programming language developed by James Gosling at Sun Microsystems in 1995.



- **Simple** – Java is easy to learn with a clean and clear syntax.
- **Object-Oriented** – Everything in Java is treated as an object, supporting concepts like inheritance, polymorphism, and encapsulation.
- **Platform Independent** – Programs run on any system having the JVM (Write Once, Run Anywhere).
- **Secure** – Java provides security through bytecode verification, exception handling, and restricted access.
- **Robust** – Java has strong memory management, garbage collection, and error handling.
- **Portable** – Java applications are portable across different platforms and operating systems.
- **High Performance** – Java uses Just-In-Time (JIT) compilation to improve execution speed.
- **Multithreaded** – Java allows multiple threads to run simultaneously for better performance.
- **Distributed** – Java supports networking and distributed computing (e.g., RMI, sockets).

How to install the Setup of Eclipse IDE :-

- Install JDK (prerequisite).
- Download Eclipse IDE from the official website (choose OS & bits).
- Install by following the standard setup process.
- Select the required developer package.
- Launch Eclipse and set a workspace folder.
- Create a new Java project and add classes.
- Write and run Java programs.